TOPIC 14

Multiply Greater Numbers

OVERVIEW

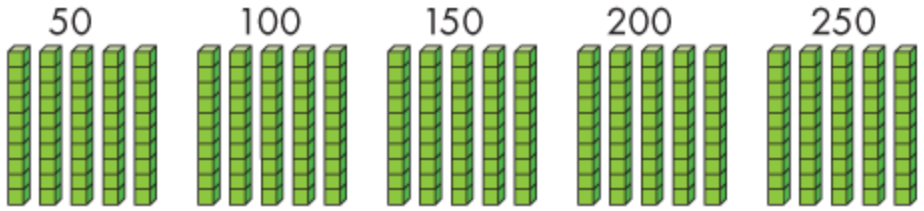
Topic 14 focuses on using place-value patterns and properties of operations to multiply greater numbers.

USE PLACE VALUE AND PATTERNS TO MULTIPLY

Place-Value Blocks and Number Lines Your child will use place-value blocks to easily see patterns when multiplying by multiples of 10. Making jumps on an open number line also reveals patterns.

Here are two ways your child can find 5×50.

Using place-value blocks and recording their value shows a pattern.



1×50 is 1 group of 5 tens = 5 tens or 50

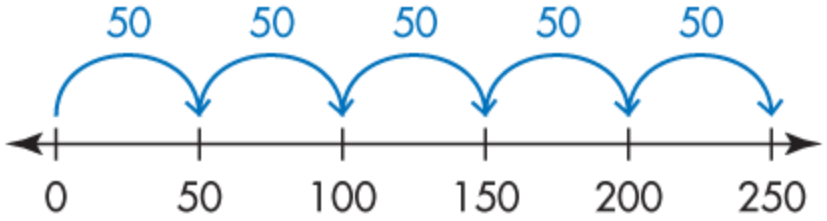
2×50 is 2 groups of 5 tens = 10 tens or 100

3×50 is 3 groups of 5 tens = 15 tens or 150

4×50 is 4 groups of 5 tens = 20 tens or 200

5×50 is 5 groups of 5 tens = 25 tens or 250

Jumps of 50 on an open number line also reveal a pattern.



1 jump of 50 is 50. $1 \times 50 = 50$

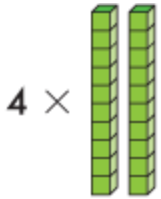
2 jumps of 50 are 100. $2 \times 50 = 100$

3 jumps of 50 are 150. $3 \times 50 = 150$

4 jumps of 50 are 200. $4 \times 50 = 200$

Mental Math Your child will use basic multiplication facts and place value to mentally multiply by multiples of 10. One goal of this topic is to practice multiplying with tens and hundreds for procedural reliability, or computing accurately and efficiently with confidence.

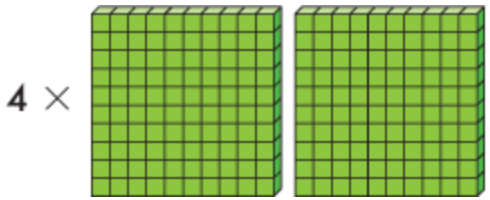
Find 4×20.



Think of 20 as 2 tens.

$4 \times 20 = 4 \times 2$ tens
$4 \times 20 = 8$ tens
$4 \times 20 = 80$

Find 4×200.



Think of 200 as 2 hundreds.

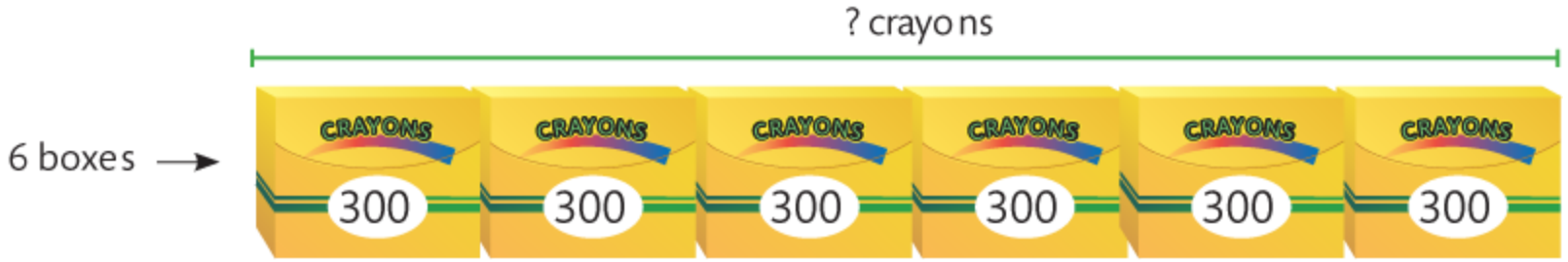
$4 \times 200 = 4 \times 2$ hundreds
$4 \times 200 = 8$ hundreds

$4 \times 200 = 800$

USE PROPERTIES TO MULTIPLY

Associative Property Your child will use the Associative Property to group numbers in different ways to multiply. By flexibly grouping numbers, your child can use basic multiplication facts to multiply greater numbers.

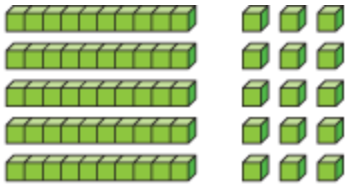
Here is how your child might solve the problem, "How many crayons are in 6 jumbo boxes of 300?"



$6 \times 300 = 6 \times (3 \times 100)$
$6 \times 300 = (6 \times 3) \times 100$
$6 \times 300 = 18 \times 100$
$6 \times 300 = 1{,}800$
There are 1,800 crayons in 6 jumbo boxes.

Distributive Property Your child will use the Distributive Property to decompose a factor. This approach will enable your child to apply previous work with the 10s facts.

Find 5×13.



Think of 13 as $10 + 3$.
$5 \times 13 = (5 \times 10) + (5 \times 3)$

$$
\begin{array}{r}
13 \\
\times \quad 5 \\
\hline
15 \quad 5 \times 3 \text{ ones} \\
+ 50 \quad 5 \times 1 \text{ ten} \\
\hline
65
\end{array}
$$

Reasonableness of Solutions Your child will consider the context of a problem and determine whether a solution is reasonable within the context.

Here is how your child might solve the problem, "A puppet theater has 6 shows a week. 72 tickets are sold for each show. How many tickets do they sell each week?"

First, estimate an answer:

$$6 \times 72 = ?$$

$$\text{Estimate: } 6 \times 70 = 6 \times 7 \text{ tens} = 42 \text{ tens} = 420$$

Then solve using partial products.

$$\begin{array}{r} 72 \\ \times \quad 6 \\ \hline 12 \\ +\ 420 \\ \hline 432 \end{array}$$

After calculating an answer, check for reasonableness:

432 is close to the estimate of 420. 432 makes sense for one week's worth of tickets for a small theater. So, the solution is reasonable.

CONNECT THE MATH

You can connect the math in this topic to everyday experiences. Involve your child in opportunities to multiply greater numbers, such as multiples of 10 and 100. Buying packs of things that come in multiples of 10 or 100 is a context for practice. For example, if you are buying 4 packages of paper plates that come in packs of 20, have your child figure out how many individual plates you will have.

TOPIC 14 LESSONS

Lesson 14-1 Use Patterns to Multiply

Lesson 14-2 Use Place Value to Multiply with Tens and Hundreds

Lesson 14-3 **PROCEDURAL RELIABILITY PRACTICE**
Multiply with Tens and Hundreds

Lesson 14-4 Use Place Value to Multiply Greater Numbers

Lesson 14-5 Use the Distributive Property to Multiply Greater Numbers

Lesson 14-6 **MATH THINKING AND REASONING**
Assess the Reasonableness of Solutions